

Olevel economics

Chp:- Demand and Supply

Topic:- Calculation of PED

A stylized handwritten signature in black ink, consisting of a large 'A' shape with a horizontal line and a loop at the bottom.

Zarale
Mushtaq

Calculation of Price Elasticity of demand

Formula:-

Price Elasticity of demand :- $\frac{\text{percentage change in quantity demand}}{\text{percentage change in price}}$

$$\% \Delta (\text{percentage change}) :- \frac{\text{New} - \text{old}}{\text{old}} \times 100$$

$$\text{percentage change in quantity demand : } \frac{\text{New } Q - \text{old } Q}{\text{old } Q} \times 100$$

(% Δ in QD)

$$\text{percentage change in price : } \frac{\text{New } P - \text{old } P}{\text{old } P} \times 100$$

(% Δ in Price)

Case #1 :-

	Price	QD
old	10	100
New	20	50

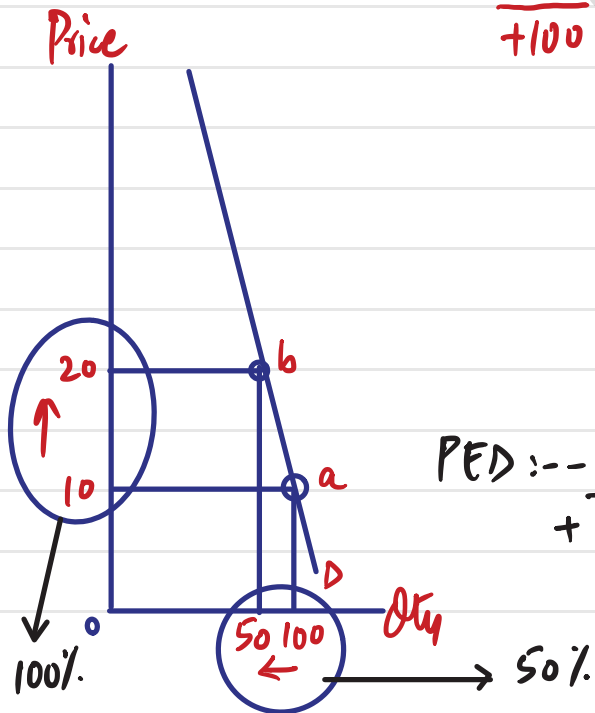
PED: ?

$$\% \Delta \text{ in } QD :- \frac{\text{New } Q - \text{old } Q}{\text{old } Q} \times 100 :- \frac{50 - 100}{100} \times 100 \Rightarrow -50$$

$$\% \Delta \text{ in price} :- \frac{\text{New } P - \text{old } P}{\text{old } P} \times 100 :- \frac{20 - 10}{10} \times 100 \Rightarrow +100$$

$$PED :- \frac{-50}{+100} \Rightarrow -0.5 (< 1) \text{ Inelastic}$$

This sign represents negative relationship between price and quantity demand which is always negative, that price and quantity demand goes in opposite direction.



$$PED :- \frac{-50\%}{+100\%} \Rightarrow -0.5$$

Case #2

old
new

Price	QD
↓ 10 5	↑ 100 200

$$\% \Delta \text{ in } QD \Rightarrow \frac{\text{New } Q - \text{old } Q}{\text{old } Q} \times 100 \Rightarrow \frac{200 - 100}{100} \times 100 \Rightarrow +100\%$$

$$\% \Delta \text{ in price} \Rightarrow \frac{\text{New } P - \text{old } P}{\text{old } P} \times 100 \Rightarrow \frac{5 - 10}{10} \times 100 \Rightarrow -50\%$$

$$PED:- \frac{\% \Delta \text{ in } QD}{\% \Delta \text{ in price}} \Rightarrow \frac{+100\%}{-50\%} \Rightarrow -2 (>1 \text{ Elastic})$$

This sign represents negative relationship between price and quantity demand. Always ignore this sign to identify the elasticity, else every answer will be considered inelastic by you!



$$PED: \frac{+100\%}{-50\%} = -2 \text{ (Elastic demand)}$$

case #3

old
new

Price	QD
10	100
↑ 15	100

→ PED ?

→ PED = 0

Perfectly inelastic demand

$$\% \Delta := \frac{\text{New} - \text{old}}{\text{old}} \times 100$$

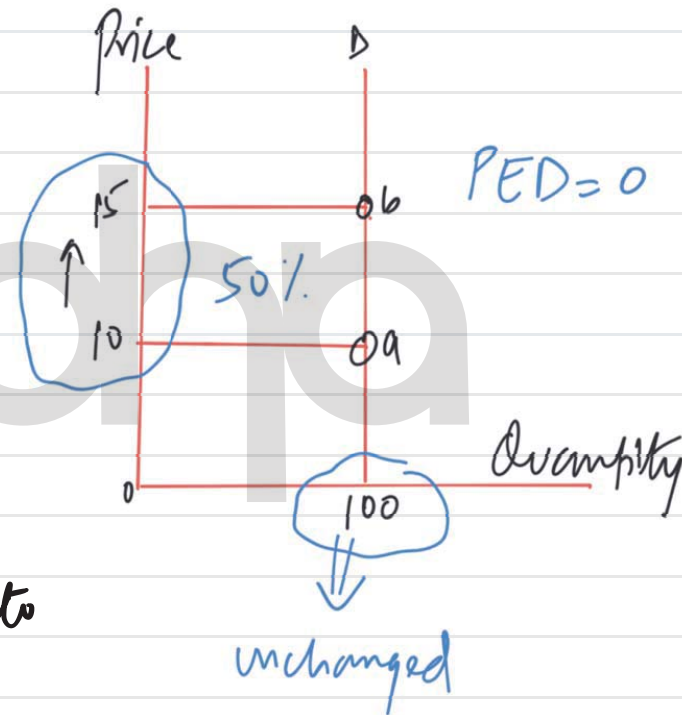
$$\% \Delta \text{ in QD} := \frac{100 - 100}{100} \times 100 = 0$$

$$\% \Delta \text{ in price} := \frac{15 - 10}{10} \times 100 = 50\%$$

$$\text{PED} := \frac{0}{50\%} \Rightarrow 0 \quad (\text{PED} = 0)$$

* Zero divided by anything is zero

* Quantity unchanged was a clear hint to identify the elasticity of demand



case #4

old
new

Price	QD
10	100
10	150

PED = ?

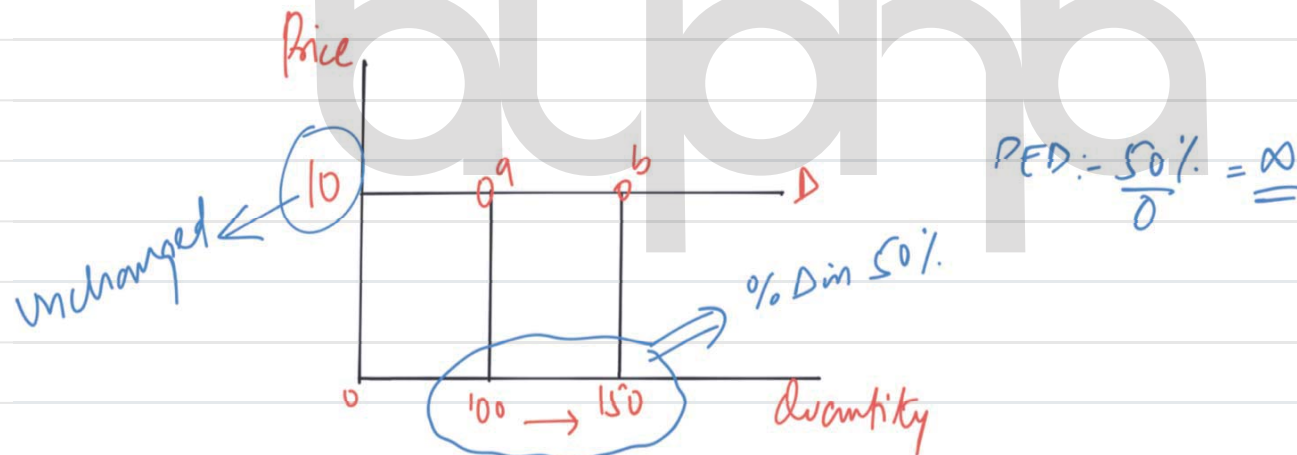
$$\% \Delta := \frac{\text{New} - \text{old}}{\text{old}} \times 100$$

Price unchanged
perfectly elastic demand
 PED = ∞

$$\% \Delta \text{ in QD} := \frac{150 - 100}{100} \times 100 = 50\%$$

$$\% \Delta \text{ in Price} := \frac{10 - 10}{10} \times 100 = 0$$

$$\text{PED} := \frac{50}{0} \Rightarrow \text{Math error, infinite } \infty, \text{ PED} = \infty$$



* Anything divided by zero will result in infinity

* Unchanged price level was a clear hint to identify the elasticity of demand.

case #5:-

Old
New

Price	QD
↓ 10 5	↑ 100 150

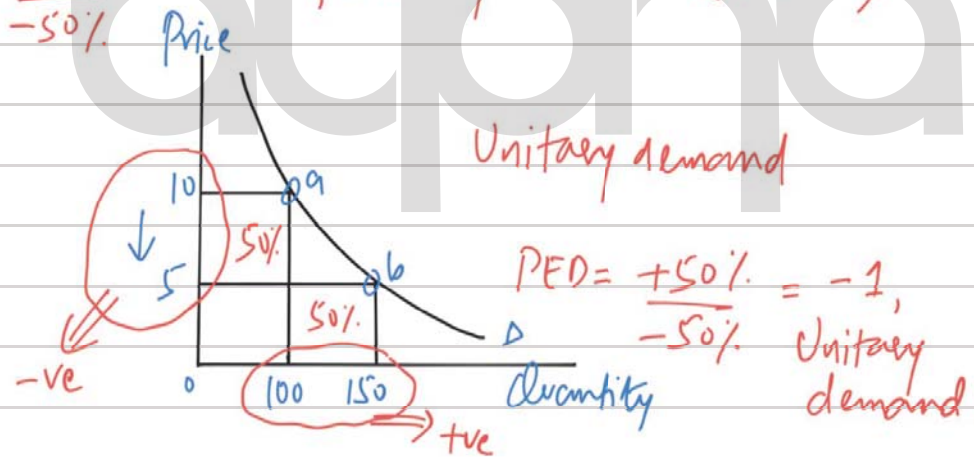
PED:-?

$$\% \Delta := \frac{\text{New} - \text{old}}{\text{old}} \times 100$$

$$\% \Delta \text{ in QD} := \frac{150 - 100}{100} \times 100 = +50\%$$

$$\% \Delta \text{ in Price} := \frac{5 - 10}{10} \times 100 = -50\%$$

$$\text{PED} := \frac{+50\%}{-50\%} = -1, \text{ Unitary demand (PED} = -1)$$



Unitary demand, a change in price brings an equal percentage change in quantity demanded. For e.g. a rise in price by 50% will contract quantity demanded by 50%, keeping total revenue or total expenditure unchanged.

Question

Q1 Using a diagram and a numerical example, explain the importance of negative sign under price elasticity of demand? 6

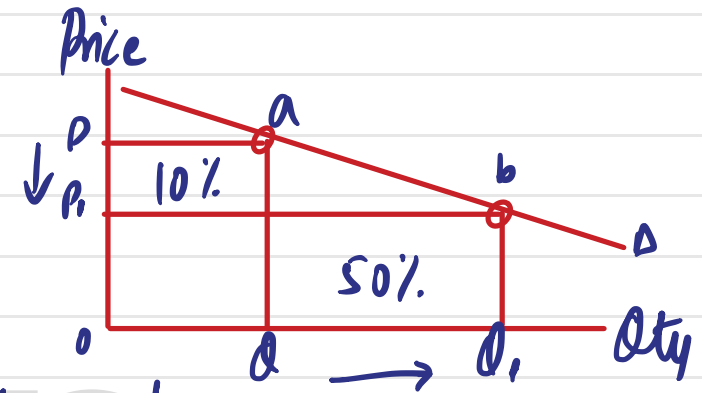
Ans:- ★ Define PED

★ formula

★ values :- $>1, <1, =1, 0, \infty$

★ Take numerical e.g. $\Rightarrow$ calculation of elastic demand
Diagram

★ Clearly mention to ignore negative sign.
(paragraph given)



$$PED := \frac{+50}{-10} = -5$$

Reason behind choosing elastic case, the negative sign under its calculation makes student feels it being less than 1, once sign is ignored, it is greater than 1.